
Abstract

Modern language agents must operate over long-horizon, multi-turn interactions, where they retrieve external information, adapt to observations, and answer interdependent queries. Yet, most LLM systems rely on full-context prompting, appending all past turns regardless of their relevance. This leads to unbounded memory growth, increased computational costs, and degraded reasoning performance on out-of-distribution input lengths. We introduce **MEM1**, an end-to-end reinforcement learning framework that enables agents to operate with **constant memory** across long multi-turn tasks. At each turn, MEM1 updates a **compact shared internal state** that jointly supports memory consolidation and reasoning. This state integrates prior memory with new observations from the environment while strategically discarding irrelevant or redundant information. To support training in more realistic and compositional settings, we propose a simple yet effective and scalable approach to constructing multi-turn environments by composing existing datasets into arbitrarily complex task sequences. Experiments across three domains, including internal retrieval QA, open-domain web QA, and multi-turn web shopping, show that MEM1-7B improves performance by $3.5\times$ while reducing memory usage by $3.7\times$ compared to Qwen2.5-14B-Instruct on a 16-**objective** multi-hop QA task, and **generalizes beyond the training horizon**. Our results demonstrate the promise of reasoning-driven memory consolidation as a scalable alternative to existing solutions for training long

ce are optimized.

1 Introduction

Large language models (LLMs) have shown remarkable performance in single-turn tasks such as question answering, summarization, and code generation [7, 51, 3]. However, emerging real-world applications increasingly operate over multiple turns—searching documents, interacting with environments [70], and making decisions based on evolving external information [53]. Examples include research agents such as OpenAI and Gemini Deep Research [37, 18] that automate complex tasks by iteratively gathering information, and web-navigation agents such as OpenManus [50] and BrowserUse [34], which must complete goals across dozens of interactive turns.

Unlike traditional tasks where the input is static or self-contained, long-horizon settings often involve answering a sequence of related questions, requiring the agent to continuously retrieve new

information, revise beliefs, and adapt to evolving contexts over time. For instance, consider a research assistant tasked with “What’s the evidence for X?”. Subsequent queries like “Who published it?” require further information retrieval, while “Is the source credible?” calls for self-reflection and assessment. Each query builds on the previously collected and accumulated information. Similarly, a shopping assistant may be first asked “Which product is cheapest?”, then “What are its reviews?”, and “Is it compatible with my device?”. These interactions span multiple turns, featuring evolving contexts and compound reasoning.

In systems designed for long-horizon settings, a common approach is to append all past observations, actions, and thoughts to the context at every turn [55, 61]. This forces the model to operate with an unboundedly growing context, which introduces three key challenges. **(1) Growing inference cost and memory usage.** Transformer-based LLMs typically incur $O(N^2)$ compute cost (or $O(N)$ with Key-Value caching) and $O(N)$ memory usage as the context length N increases [52]. Consequently, deploying these models requires reserving large GPU memory on modern inference frameworks to accommodate the growing context [27, 68], often leading to significant wastage of computing resources. **(2) Generalization limits beyond the training horizon.** Ongoing conversations with context length exceeding that in the training data become out-of-distribution for the model. The model struggles to manage and reason over such unfamiliar long-horizon inputs [63]. **(3) Overloaded and inefficient context.** The accumulation of irrelevant or redundant content dilutes the model’s attention. This reduces its ability to reason effectively, even when relevant information is technically still present within the prompt [2, 30, 56].

Recent progress in long-context modeling largely targets static inputs (*e.g.*, long documents) and does not address multi-turn interaction with external environments [6, 19]. Some other approaches introduce external memory modules (*e.g.*, summarizers or retrievers) [63, 29, 13, 57], but these are typically trained separately and cannot be optimized end-to-end with the agent’s policy. This also introduces additional engineering overhead, as engineers must manage and integrate two separate models. Meanwhile, existing works on tool-using agent systems trained with reinforcement learning leave memory management unsolved, letting the prompt length grow unboundedly [24, 69]. A natural question is raised: *Can a language model learn to consolidate its memory as part of its reasoning process so that it retains only what is essential for solving the task?*

Motivated by this question, we present MEM1: **Memory-Efficient Mechanism** via learning 1-step integrated reasoning and consolidation—a method for training LLM agents that maintain constant memory usage across arbitrarily long horizons. As illustrated in Fig. 1, at each turn, the model updates a consolidated state composed of prior memory and newly obtained information. This consolidated state becomes the agent’s only retained memory, allowing all external tool outputs to be discarded after use, which prevents prompt expansion altogether as illustrated by Fig. 2. A key insight of our method is that inference-time reasoning [55, 15, 33, 65] serves a dual function: it not only provides deeper insight into the current query but also acts as a form of “working memory” [4], extracting key components from gathered information to build an evolving understanding. By unifying reasoning and memory consolidation, MEM1 offers an elegant solution where the agent learns to reason and remember within a shared representational space without requiring additional modules or architectural changes.

We train this behavior end-to-end with reinforcement learning (RL) [48, 71], optimizing for task success via verifiable rewards [43]. Although not explicitly optimized for memory efficiency through reward signals, the agent learns to manage memory as part of its policy, resulting in near-constant memory usage across long horizons. Additionally, we notice that current training and evaluation environments predominantly focus on single-objective tasks [26, 59, 38], limiting their ability to fully prepare agents for realistic, long-horizon scenarios that inherently involve multiple sequential objectives. To address this challenge, we introduce a scalable task augmentation approach, transforming existing single-objective QA datasets into complex multi-objective tasks through compositions of N multi-hop questions. This approach enables us to repurpose standard benchmarks in our community to more effectively train and evaluate agents on long-horizon reasoning, an increasingly important capability in real-world applications.

To evaluate our method comprehensively, we employ diverse multi-turn environments, including internal retrieval-augmented QA [26, 59], open-domain Web QA [69], and complex multi-turn agent shopping scenarios in WebShop [60]. Across these scenarios, MEM1 consistently rivals the performance of leading baselines while delivering efficiency gains up to $3.5\times$ in memory usage. Moreover, agents trained on our augmented 2-objective compositions generalize effectively to tasks involving up to 16-objective compositions. Notably, at the 16-objective level, our MEM1 achieves superior accuracy compared to all baseline methods, along with $1.27\times$ lower peak memory usage and $1.78\times$ faster inference compared to the respective best uncollapsed baseline.

3 MEM1

Complex reasoning tasks often require an iterative process of information gathering and synthesis, as seen in applications such as “deep research” [37, 23] and web-based agents [35, 20]. Recent advances in agent design involve interaction loops that interleave chain-of-thought reasoning [55, 15], environment interaction, and real-world feedback collection. To explicitly capture these core elements, we annotate each component using XML-style tags: `<IS>` for internal state (reasoning), `<query>` for environment queries, `<answer>` for the agent’s responses, and `<info>` for external observations or tool outputs. MEM1 adopts a **learned** approach to iterative state updating and consolidation, ensuring that only the most recent set of `<IS>`, `<query>`, `<answer>`, and `<info>` elements is retained in the prompt at any given time. This design maintains a bounded and semantically relevant context, promoting efficient and coherent multi-step reasoning.

3.1 Memory as Part of Reasoning

To achieve a constant memory, MEM1 is particularly trained to iteratively refine its understanding by processing new information in conjunction with a consolidation of its prior state. At each turn t , the agent produces a new `<IS_t>` element, which summarizes past information and reasons about subsequent actions. Following this, the agent generates an action—either a `<query_t>` to interact with the environment, or an `<answer_t>` if a direct response is warranted. If the agent issues a `<query_t>`, the corresponding feedback from the environment is appended as `<info_t>`. At the next turn, $t + 1$, the agent consolidates the tuple $(\text{<IS}_t\text{>, <query}_t\text{>, <info}_t\text{>})$ into a new `<IS_(t+1)>`, which serves as the basis for further interactions. After each turn, all tags from the previous turn t are pruned from the context, effectively compressing memory and preventing prompt bloat. Fig. 3 (bottom left) illustrates the evolution of the model’s context over time. At any given turn, the agent retains at most two `<IS>` elements, two `<query>` elements, and one `<info>` element, ensuring bounded and efficient memory usage. The detailed rollout algorithm is in Alg. 1 of App. A.5.

RL offers a powerful mechanism for shaping agent behavior through reward signals [49]. In MEM1, we leverage this framework to incentivize effective state consolidation by designing environments in which the agent is rewarded only when it strategically retains and integrates useful information. Specifically, we construct tasks that require numerous interactions with the environment to arrive at a correct answer (see Sec. 3.3). Success depends on the agent’s ability to rely on information collected along the inference path. At each turn, we prune the agent’s context to retain only the most recent `<IS>`, forcing the agent to perform memory consolidation as part of its reasoning process. Without access to full historical context, the agent must learn to preserve and update relevant knowledge internally in order to reap the reward. This learning procedure mirrors how humans

cultivate memorization skills through structured tasks such as Sudoku or crosswords [1], where success hinges on selectively attending to key information and building upon it. Over time, such tasks help individuals develop cognitive strategies that jointly support efficient memorization and reasoning, similar to our RL method for training MEM1.

3.2 Masked Trajectory for Policy Optimization

In the previous section, we detailed the rollout process of our RL pipeline. However, unlike conventional multi-turn agents that preserve a static context during generation, MEM1 introduces a unique challenge: its mechanism continuously updates the context at each turn by consolidating prior memory and pruning irrelevant tokens. This dynamic context update disrupts the continuity of the token generation trajectory, complicating the estimation of token-wise advantages in policy optimization algorithms such as PPO and Reinforce++ [41, 22], where trajectories are typically assumed to be linear.

To address this, we introduce a masked trajectory approach that reconstructs a logically coherent full trajectory by stitching together multiple interaction turns with evolving contexts. This unified trajectory mimics a standard multi-turn rollout and comprises a sequence of tuples $\tau_t = (\text{IS}_t, \text{query}_t, \text{info}_t)$ for $t \in [1, T - 1]$, where T denotes the total number of interaction turns. The T -th turn outputs the final answer $\tau_T = (\text{IS}_T, \text{answer}_T)$. The full trajectory encodes all information needed for accurate policy learning while respecting MEM1’s memory consolidation at each turn. Fig. 3 (bottom left) demonstrates the evolution of the agent’s context.

To ensure that policy gradients are correctly computed under this consolidated memory regime, we apply a **two-dimensional attention mask** [40] across the full trajectory. This mask restricts each token’s attention to only the tokens retained in memory at the time that token was generated. Specifically, for a token position k , we mask out all prior tokens that are not part of the consolidated memory corresponding to the context at that turn. This masking strategy enables accurate computation of the policy objective: letting s_k denote the masked input state and a_k the predicted token, the log-probability ratio $\rho_k(\theta) = \frac{\pi_\theta(a_k | s_k)}{\pi_{\theta_{\text{old}}}(a_k | s_k)}$ remains valid and tractable, in turn ensuring that the advantage, KL penalty, and value estimation are correct. A visualization is in Fig. 3 (bottom right). Furthermore, following [24], we incorporate a one-dimensional attention mask over retrieved external information

during policy updates for both the actor and critic networks. This ensures that gradient updates are localized to only tokens generated by the agent. Fig. 3 (bottom right) shows the masking mechanism that enables stable and accurate policy optimization under MEM1’s memory-constrained execution.

3.3 Multi-Objective Task Design

Although our proposed method is designed to address the critical challenges of agentic multi-turn interaction with the external world, there are limited publicly available datasets that support training for such long-horizon interactive processes. Existing benchmarks, such as HotpotQA [59], Bamboogle [38], and 2wiki [21], are often cited as multi-hop benchmarks, yet they typically involve only two information-seeking steps. Moreover, these datasets are not explicitly structured to support long-horizon interactions that necessitate the agent to manage the memory state.

To bridge this gap, we introduce a novel task—multi-objective question & answering (QA)—that extends the number of reasoning steps required to solve a problem. Building on existing multi-turn datasets such as HotpotQA and Natural Question [59, 26], we interleave multiple questions from the original QA corpus and construct a single composite query that requires answering all constituent sub-questions, shown in Prompt 1 of App. A.3. This formulation compels the agent to perform multiple search queries, each targeting a distinct sub-question or sub-objective, and then integrate the retrieved answers to form a comprehensive final response. Compared to the original tasks, our synthesized multi-objective, multi-hop setting significantly increases the number of search and reasoning turns required, leading to more complex, memory-intensive interactions.

4 Experiments & Results

We empirically demonstrate the effectiveness of our approach in training the MEM1 agent to perform multi-turn tasks while preserving a near-constant-sized memory state. We evaluate MEM1 against several baselines using a comprehensive set of metrics categorized into *accuracy* (e.g., Exact Match, F1 score, Environment Reward) and *efficiency* (e.g., Peak Token Usage, Dependency Length, Inference Time). All MEM1 variants are fine-tuned from the Qwen2.5-7B Base model [58]. We use PPO [41] as the RL algorithm as it computes token-level advantages, bringing stability to the training process. While we also experimented with instruction-tuned and supervised fine-tuned models using curated high-quality trajectories, reinforcement learning from the base model consistently yielded the best performance and generalization.

Our experiments are conducted in two standard environments, each reflecting real-world scenarios that require multi-turn agent interactions. The first environment is question answering with retrieval-augmented generation (RAG) [26, 59], where the agent must answer queries by retrieving relevant information from an external knowledge store (either a database or an online search engine). We trained on RAG with a local database (*i.e.*, Wikipedia Corpus) and evaluated on tasks involving open web browsing. For QA, following Sec. 3.3, we construct multi-objective tasks and tested the model performance on tasks with more questions than seen in the training. The second environment is WebShop navigation [60], where the agent assists users in online shopping by browsing a website and selecting items based on natural language descriptions. This task requires the agent to iteratively read page content and make navigation decisions, following protocols similar to those in WebGPT [35].

4.1 Implementation Details

Datasets and evaluation metrics. We train two versions of MEM1 agent for both long-horizon QA and web navigation. For long-horizon QA, we augment the multi-hop QA dataset from [24] that mixes data from both HotpotQA [59] and Natural Question [26] to form a 2-objective composite task.

For the web agent, we use the WebShop environment [60], which also produces a reward during training [64]. For all datasets, the train-test split follows the original papers. During RL training, we employ the exact match (EM) metric for QA tasks (details in App. A.4.1) and the environment reward for WebShop [60, 64]. To evaluate the effectiveness of various approaches, we measure the EM and F1 score for QA tasks and final reward for the WebShop environment [60, 64]. To evaluate the efficiency, we consider the peak token usage, average dependency, and average inference time. The test datasets are obtained from the original papers which consist of out-of-distribution data. The former two metrics measure the memory efficiency, while the latter measures the time efficiency.

The detailed definitions of the metrics are in App. A.4.1. The prompt and format can be found in App. A.3.

Baselines. To evaluate the accuracy and efficiency of MEM1, we compare it against a diverse set of baselines designed to either enhance task performance or manage context effectively. For the QA environment, we benchmark accuracy against Search-R1 [24], DeepResearcher [69], and a larger-scale model, Qwen2.5-14B-Instruct [58]. Details about Search-R1 and DeepResearcher can be found in App. A.4.2. For the WebShop environment, we compare against Agent-FLAN [11], Agent-R [64], and AgentLM [66]. To assess efficiency, we consider two context compression baselines using models of the same parameter size as MEM1. First, we apply MEM1’s agentic *truncation* prompt template and rollout to a standard instruct model, isolating the benefits of prompt and rollout design alone. Second, we evaluate A-MEM [57], which augments an Instruct model with a vector database for memory retrieval, capturing the effect of external memory modules in agentic systems. We additionally train a supervised fine-tuned (SFT) model using trajectories curated from GPT-4o [36] based on MEM1’s rollout and compare it with the RL-trained agent.

Meta info injection. In our agentic pipeline, the agent’s context is programmatically truncated at each turn—immediately after it generates a search query or an answer—following the procedure outlined in Sec. 3. As past context is truncated, the agent may have difficulty determining when to terminate. To address this, we prepend a hint [HINT: YOU HAVE {turns_left} TURNS LEFT] at the beginning of each `<info>` tag to remind the agent of its remaining turns budget. For all experiments, we set the maximally allowed turns to 6 for 1-objective to 4-objective tasks and 20 for more difficult tasks to avoid excessively long trajectories.

4.2 MEM1 on Multi-Objective Multi-Hop Tasks

One key advantage of MEM1 agents lies in their efficient management of long-horizon interactions with the environment. To demonstrate this, we train our MEM1 agent with a 2-objective augmentation of the QA dataset, and subsequently test it against other models, using held-out multi-objective test datasets similarly augmented from the original test datasets. As elaborated in Sec. 3.3, these multi-objective tasks require a significantly larger number of turns of environment interactions to complete, hence serving as better benchmarks for memory management. As shown in Tab. 1, when evaluated on 2-objective datasets, MEM1 achieves better performance (in terms of EM and F1 scores) than other 7B counterparts, while incurring significantly lower peak token usage and achieving faster inference time.

The advantage of MEM1 becomes even more evident in tasks requiring longer-horizon interactive processes. To highlight such scalability of MEM1, we further compare the models on 3, 4, 6, 8, and 16-objective tasks in Fig. 4 and Tab. 1. Fig. 4 illustrates the scaling trends of task performance (measured by EM count) and memory efficiency (measured by Peak Token Usage) for MEM1 relative to other models and memory management baselines. As the number of objectives increases, the Peak Token Usage of all other methods and models scales nearly linearly. In contrast, MEM1 maintains an almost constant peak token count with only a slight increase, as also shown in Tab. 1.

Notably, while MEM1 initially underperforms Qwen2.5-14B-Instruct, its performance gradually catches up as the number of objectives increases, eventually surpassing the 14B model, which has double the parameter count. MEM1 also demonstrates remarkable efficiency. In the 16-objective task, it requires only 27.1% of the peak tokens and 29.3% of the total inference time compared to Qwen2.5-14B-Instruct. This efficiency translates to significantly reduced GPU memory requirements and overall computing resource demands.

4.3 MEM1 on Single-Objective Multi-Hop Tasks

While MEM1 is designed to train agents for very long-horizon tasks, our training method also delivers improved capability with existing multi-hop tasks while achieving much greater efficiency at the same time, all without being explicitly trained on the single-objective versions of these tasks. Note that single-objective tasks also require multiple turns of interaction to produce the desired output.

Long-horizon web navigation in WebShop. Beyond QA tasks, we further evaluate the effectiveness of MEM1 in managing long-horizon interactions in the form of web navigation. We show

the experimental results in Tab. 2. Trained in the WebShop environment (see App. A.6), MEM1 outperforms other agent training baselines, including Agent-Flan, Agent-R, and AgentLM when utilizing models of similar size. Furthermore, MEM1 achieves remarkable efficiency improvements compared to the best baseline method, AgentLM, featuring a $2.8\times$ improvement in Peak Token Usage, a $1.9\times$ improvement in Dependency, and a $1.5\times$ improvement in Inference Time. MEM1 even surpasses AgentLM-13B, a model with twice the parameter count of our trained model. Additionally, our results indicate that using MEM1 is significantly better than OpenAI’s GPT-4o on the WebShop tasks, even when the truncation prompt templates or A-MEM techniques are applied to GPT-4o.

Single-objective QA in Wikipedia. Tab. 3 presents the accuracy and efficiency metrics for evaluations on single-objective QA tasks on Wikipedia [24], where the agent can make retrieval requests from the Wikipedia datastore via RAG. The MEM1 used in this evaluation is the same as the one detailed in Sec. 4.2, which is trained solely on a 2-objective task. Overall, MEM1 demonstrates superior efficiency across all three evaluated efficiency metrics, while simultaneously achieving the highest EM score and an F1 score comparable to that of Qwen2.5-14B-Instruct. This improvement in efficiency is attributed to the MEM1 agent’s ability to consolidate memory from previous interactions into a compact internal state, which reduces the number of tokens used in the context. We also observe that SFT significantly underperforms RL, highlighting the necessity for RL-based training.

Zero-shot transfer to Online Web-QA. To validate the transferability and generalizability of the trained MEM1 agent, we perform a zero-shot transfer to an online web-QA environment, which is unseen by the agent. In this environment, agents conduct web searches through an API service that returns results including titles, snippets, and URLs. As shown in Tab. 3, MEM1 consistently exhibited improved efficiency alongside comparable effectiveness in this unseen setting via zero-shot transfer.

4.4 Analysis on Emergent Agent Behaviors

Through analyzing MEM1’s multi-turn interaction traces trained on 2-objective QA, we observe a range of emergent behaviors that are critical for handling long-horizon, multi-objective tasks, demonstrating capabilities well beyond simple retrieval. First, MEM1 learns to **manage multiple questions concurrently** by maintaining a structured internal state. As shown in Fig. 5(a), when faced with two multi-hop questions, the agent stores and updates memory for each question separately, guiding subsequent searches based on the identified information gaps. In (b), MEM1 exhibits the ability to shift focus when progress on one question stalls, recognizing difficulty and prioritizing the more tractable objective. Meanwhile, MEM1 learns to **interleave reasoning and memory** in its internal state `<IS>`, weaving important information into its decision-making process to support both information retention and action selection. In Fig. 5 (c), MEM1 explicitly extracts important information from previous search results and leverages it to formulate the next query that best addresses the current information gap. In addition, (d) shows that when new, relevant information is retrieved, MEM1 explicitly reasons about its significance and selectively updates its memory. We believe that learning these interleaved behaviors is key to achieving efficiency gains in memory without degrading performance. Beyond behaviors unique to our multi-objective setup and memory architecture, MEM1 also exhibits several **general-purpose search strategies**. In (e), the agent performs self-verification, correcting an earlier misconception and issuing a new query for confirmation. In (f), complex queries are decomposed into manageable subgoals before initiating the search. In (g), for questions requiring multi-turn information gathering, MEM1 extracts key information from search results and uses it to inform the next search. In (h), when overly specific queries fail, MEM1 re-scopes its query to improve retrieval. Notably, many of these behaviors, including verification, making a plan, and iterative search, are also reported in recent studies on deep research agents [24, 69].

5 Conclusion, Limitations, and Future Work

We introduced MEM1, a reinforcement learning framework that enables language agents to perform long-horizon reasoning with consolidated memory. By integrating inference-time reasoning and memory consolidation into a unified internal state, MEM1 addresses the scalability challenges of prompt growth and achieves competitive performance across QA and web navigation benchmarks, with substantially reduced memory usage and inference latency. Despite these advantages, MEM1 assumes access to environments with well-defined and verifiable rewards. While this assumption holds in domains such as QA, math, and web navigation, many open-ended tasks present ambiguous or noisy reward structures. Fully realizing the potential of MEM1 therefore requires advances in modeling such tasks and designing suitable reward mechanisms—challenges that lie beyond the scope of this work. A promising future direction is to explore methods for training MEM1 agents in these open-ended settings where reward signals are sparse, delayed, or implicit.